Rewarding Learning

ADVANCED
General Certificate of Education
2018

Centre Number

Candidate Number

Digital Technology

Assessment Unit A2 1

assessing

Information Systems

[CODE]

SPECIMEN PAPER

TIME

2 hours 30 minutes.

INSTRUCTIONS TO CANDIDATES

Write your Centre Number and Candidate Number in the spaces provided at the top of this page.
Write your answers in the spaces provided in this question paper.
Answer **all seven** questions.

INFORMATION FOR CANDIDATES

The total mark for this paper is 150.
Figures in brackets printed down the right-hand side of pages indicate the marks awarded to each question or part question.
Quality of written communication will be assessed in questions **3(c), 5(c), 6(b)** and **6(c)**.

For Examiner's use only	
Question Number	**Marks**
1	
2	
3	
4	
5	
6	
7	
Total Marks	

Version 2: 6 March 2017

Examiner Only	
Marks	Re-mark

1 **(a)** Describe the purpose of each of the following network resources:

Server

_____

_____

_____

_____ [2]

Wireless access point

_____

_____

_____

_____ [2]

MAC address

_____

_____

_____

_____ [2]

(b) Compare and contrast a peer to peer network with a server based network.

[4]

Ring and star are two types of network topology.

(c) Describe the main features of a ring network.

[3]

Examiner Only	
Marks	Re-mark

(d) Compare the ring topology with the star topology with respect to each of the following:

Security

[3]

Reliability

[3]

(e) The OSI network model consists of a number of layers.

Describe the main features of the physical layer.

[3]

Examiner Only	
Marks	Re-mark

(f) Compare metal cable and fibre optic cable with respect to each of the following:

Examiner Only	
Marks	Re-mark

Bandwidth

[3]

Security

[3]

Examiner Only	
Marks	Re-mark

2 A library stores details of its members, books, and loans in a relational database.

(a) Describe the main features of a relational database.

[4]

The database includes the following tables:

- **Book** (BookID, Title, Author, Publisher, Category)
- **Member** (MemberID, Surname, Forename, HouseNumber, Postcode, EmailAddress)
- **Loan** (MemberID, BookID, DateOfLoan, DateDueBack, Returned)

(b) Using examples from this database, explain each of the following terms:

Entity

[2]

Attribute

[2]

Primary key

_____

_____

_____

_____ [2]

Composite key

_____

_____

_____

_____ [2]

Examiner Only	
Marks	Re-mark

(c) Show how the Book, Member and Loan tables could be represented in an Entity Relationship (ER) Diagram.

[4]

(d) Data in the library's database has been normalised.

Describe **two** advantages of data which is normalised.

1 _____

_____

_____

_____ [2]

2 _____

_____

_____

_____ [2]

Examiner Only	
Marks	Re-mark

Structured Query Language (SQL) is used to

- maintain data in the database
- create queries

Examiner Only	
Marks	Re-mark

(e) Using SQL produce a command which will add details of the following loan to the database.

MemberID:	43216
BookID:	21345
DateOfLoan:	24/08/2016
DateDueBack:	31/08/2016
Returned:	False

[3]

(f) Using SQL produce a query which will retrieve the Title and Author of all Science books in the database.

[3]

(g) The librarian is not a Digital Technology expert.

Explain why the librarian might use Query by Example (QBE) to create a query.

_____

_____

_____

_____

_____

_____

_____

_____ [4]

Examiner Only	
Marks	Re-mark

3 Digital Technology plays a key role in mobile phone communication.

(a) Describe the technology which enables a mobile phone user to make phone calls while on the move.

[6]

Examiner Only	
Marks	Re-mark

A hospital uses an expert system to help diagnose patients' illnesses before they are seen by a consultant.

Examiner Only	
Marks	Re-mark

(b) Explain the meaning of each of the following terms:

Expert system shell

[3]

Fuzzy logic

[3]

Heuristics

[3]

(c) Evaluate the hospital's use of an expert system.

Quality of written communication will be assessed in this question

Examiner Only	
Marks	Re-mark

[9]

Examiner Only	
Marks	Re-mark

4 Data mining is used by many businesses to increase their competitiveness.

(a) What is meant by data mining?

[4]

(b) Explain how a chain of supermarkets could use data mining to increase its competitiveness.

[7]

Examiner Only	
Marks	Re-mark

5 When using Personal Data most organisations must comply with the Data Protection Act.

(a) Identify **four** principles specified in the Data Protection Act.

1 _____

_____

2 _____

_____

3 _____

_____

4 _____

_____ [4]

(b) Describe **two** rights the DPA gives the data subject.

1 _____

_____

_____

_____ [2]

2 _____

_____

_____

_____ [2]

(c) Discuss the main ethical considerations relating to the use of personal information by companies.

Quality of written communication will be assessed in this question

[9]

Examiner Only	
Marks	Re-mark

6 Many organisations make use of virtualisation and cloud computing.

(a) Describe how virtualisation can be used on a computer system.

[6]

Examiner Only	
Marks	Re-mark

A large organisation decides to upgrade its payroll system.
The organisation decides to use a third party to host the new payroll system.

(b) Evaluate the organisation's decision to use a third party with respect to security.

Quality of written communication will be assessed in this question

[9]

Examiner Only	
Marks	Re-mark

(c) Explain the impact of the increasing use of cloud computing on the memory requirements of personal computers.

Quality of written communication will be assessed in this question

[9]

Examiner Only	
Marks	Re-mark

7 When data is sent over a network it is important that data transmission errors are detected.

Examiner Only	
Marks	Re-mark

(a) Describe how each of the following can be used to detect data transmission errors:

Parity bits

[4]

CRC

[4]

When sensitive data is sent over the Internet it is important that it is kept secure.

Examiner Only	
Marks	Re-mark

(b) Explain how each of the following can be used to keep data secure:

SSL

[4]

Digital Certificate

[4]